

AI5707 – Pattern Recognition and Machine Learning

Assignment 5: Maximum Likelihood Parameter Estimation

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Dataset clarification

Question 1 is the synthetic Gaussian experiment. Question 2 uses the uploaded real-world train/dev data. The report includes confusion matrices for both questions.

1 Question 1: Synthetic Gaussian Data

1.1 Method

Generate 10,000 points per class, with means $(-4, -1.5)$, $(0, 4)$, and $(4, -1)$. Compare isotropic ($\sigma^2 I$), diagonal, and full covariance structures with shared and class-specific covariance. Use an 80:20 train-test split.

$$\hat{\mu}_i = \frac{1}{n_i} \sum_{k=1}^{n_i} x_k, \quad \hat{\Sigma}_i = \frac{1}{n_i} \sum_{k=1}^{n_i} (x_k - \hat{\mu}_i)(x_k - \hat{\mu}_i)^T.$$

1.2 Results summary

Generated covariance	Setting	Best model	Accuracy
Isotropic	Shared covariance	Isotropic	99.9333%
Isotropic	Class-specific covariance	Isotropic	99.8667%
Diagonal	Shared covariance	Diagonal	99.9833%
Diagonal	Class-specific covariance	Diagonal	99.9833%
Full	Shared covariance	Diagonal	99.4500%
Full	Class-specific covariance	Full	99.9167%

1.3 All 18 confusion matrices

Rows are actual classes; columns are predicted classes.

Table 1: Isotropic generated; Shared covariance; fitted Isotropic; accuracy 99.9333%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	1997	3
Class 3	0	1	1999

Table 2: Isotropic generated; Shared covariance; fitted Diagonal; accuracy 99.9333%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	1997	3
Class 3	0	1	1999

Table 3: Isotropic generated; Shared covariance; fitted Full; accuracy 99.9333%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	1997	3
Class 3	0	1	1999

Table 4: Isotropic generated; Class-specific covariance; fitted Isotropic; accuracy 99.8667%

Actual \ Predicted	1	2	3
Class 1	1996	4	0
Class 2	3	1996	1
Class 3	0	0	2000

Table 5: Isotropic generated; Class-specific covariance; fitted Diagonal; accuracy 99.8667%

Actual \ Predicted	1	2	3
Class 1	1996	4	0
Class 2	3	1996	1
Class 3	0	0	2000

Table 6: Isotropic generated; Class-specific covariance; fitted Full; accuracy 99.8667%

Actual \ Predicted	1	2	3
Class 1	1996	4	0
Class 2	3	1996	1
Class 3	0	0	2000

Table 7: Diagonal generated; Shared covariance; fitted Isotropic; accuracy 99.8333%

Actual \ Predicted	1	2	3
Class 1	1999	1	0
Class 2	2	1992	6
Class 3	0	1	1999

Table 8: Diagonal generated; Shared covariance; fitted Diagonal; accuracy 99.9833%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	2000	0
Class 3	0	1	1999

Table 9: Diagonal generated; Shared covariance; fitted Full; accuracy 99.9833%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	2000	0
Class 3	0	1	1999

Table 10: Diagonal generated; Class-specific covariance; fitted Isotropic; accuracy 99.5333%

Actual \ Predicted	1	2	3
Class 1	1996	4	0
Class 2	8	1986	6
Class 3	0	10	1990

Table 11: Diagonal generated; Class-specific covariance; fitted Diagonal; accuracy 99.9833%

Actual \ Predicted	1	2	3
Class 1	1999	1	0
Class 2	0	2000	0
Class 3	0	0	2000

Table 12: Diagonal generated; Class-specific covariance; fitted Full; accuracy 99.9833%

Actual \ Predicted	1	2	3
Class 1	1999	1	0
Class 2	0	2000	0
Class 3	0	0	2000

Table 13: Full generated; Shared covariance; fitted Isotropic; accuracy 99.4333%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	1982	18
Class 3	0	16	1984

Table 14: Full generated; Shared covariance; fitted Diagonal; accuracy 99.4500%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	1986	14
Class 3	0	19	1981

Table 15: Full generated; Shared covariance; fitted Full; accuracy 99.4167%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	0	1984	16
Class 3	0	19	1981

Table 16: Full generated; Class-specific covariance; fitted Isotropic; accuracy 99.6500%

Actual \ Predicted	1	2	3
Class 1	1996	4	0
Class 2	0	1983	17
Class 3	0	0	2000

Table 17: Full generated; Class-specific covariance; fitted Diagonal; accuracy 99.6500%

Actual \ Predicted	1	2	3
Class 1	1997	3	0
Class 2	0	1982	18
Class 3	0	0	2000

Table 18: Full generated; Class-specific covariance; fitted Full; accuracy 99.9167%

Actual \ Predicted	1	2	3
Class 1	2000	0	0
Class 2	1	1996	3
Class 3	0	1	1999

1.4 Decision-boundary plots

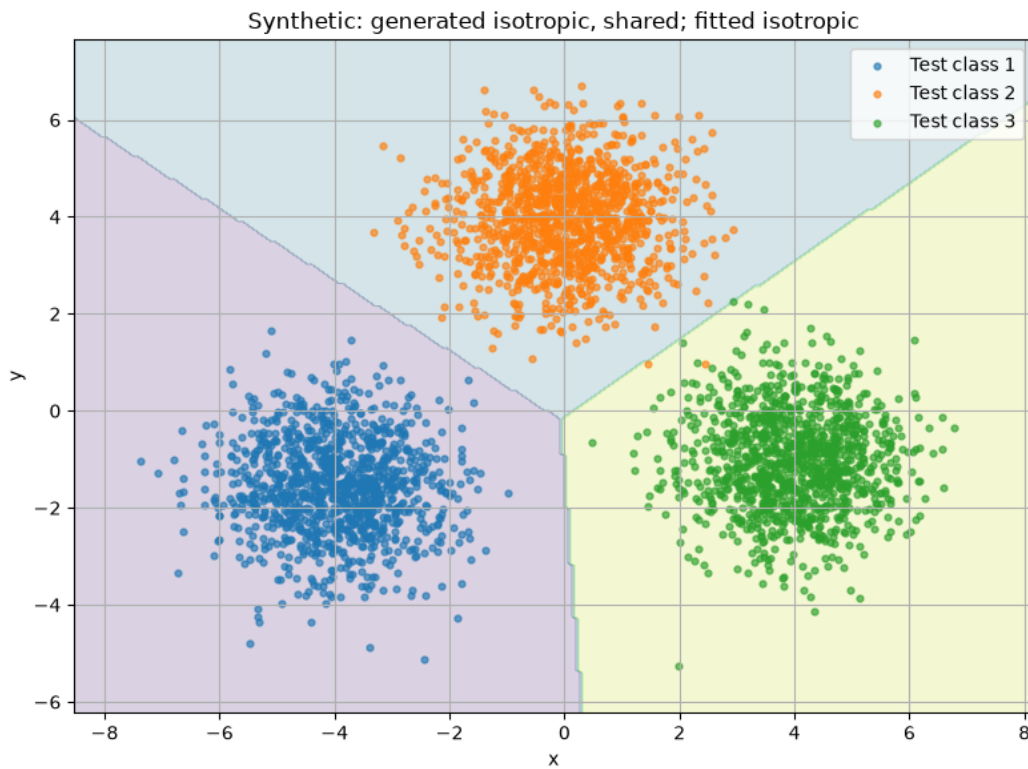


Figure 1: Q1 Figure 1: Confusion matrix plot

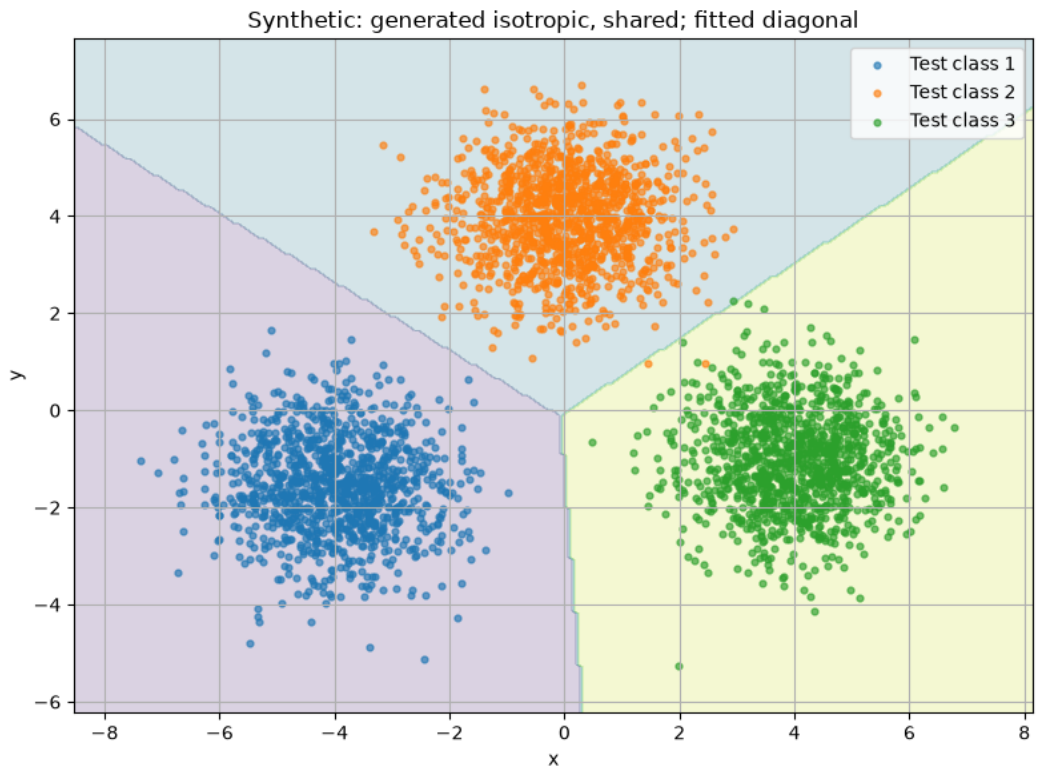


Figure 2: Q1 Figure 2: Confusion matrix plot

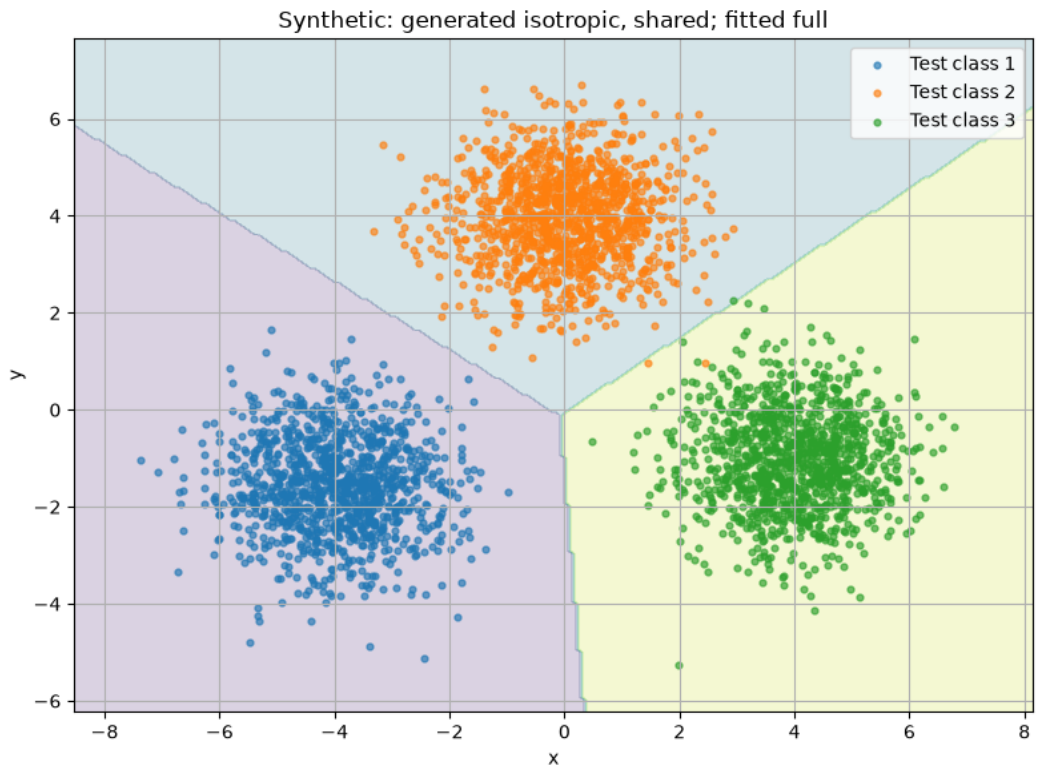
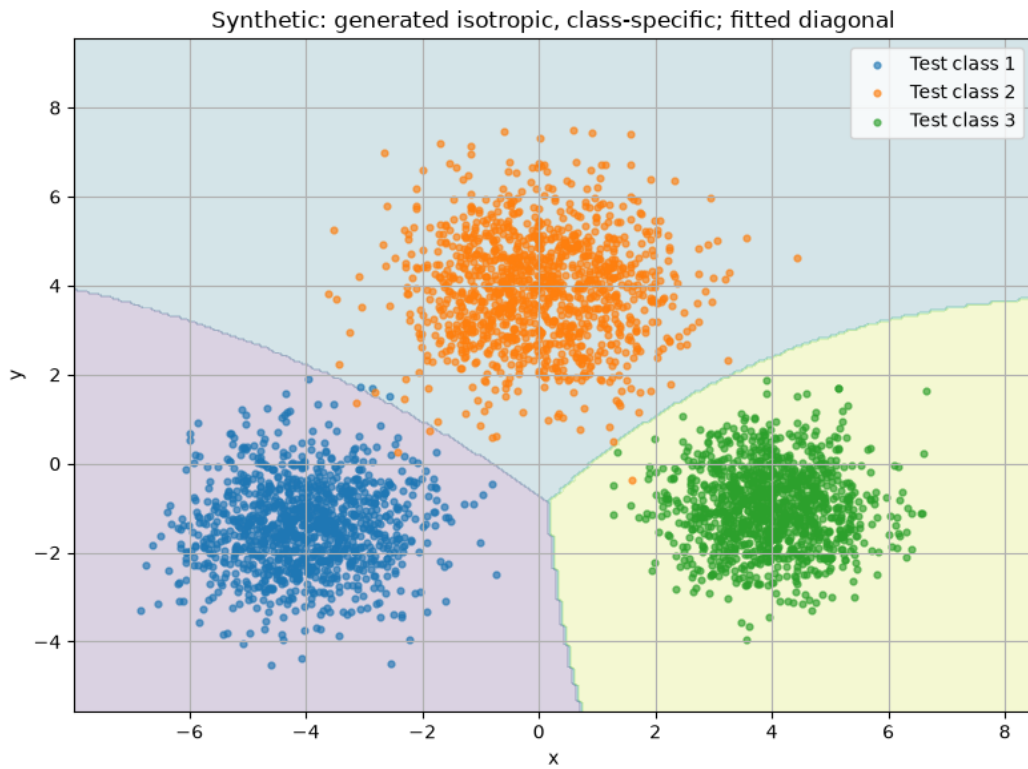
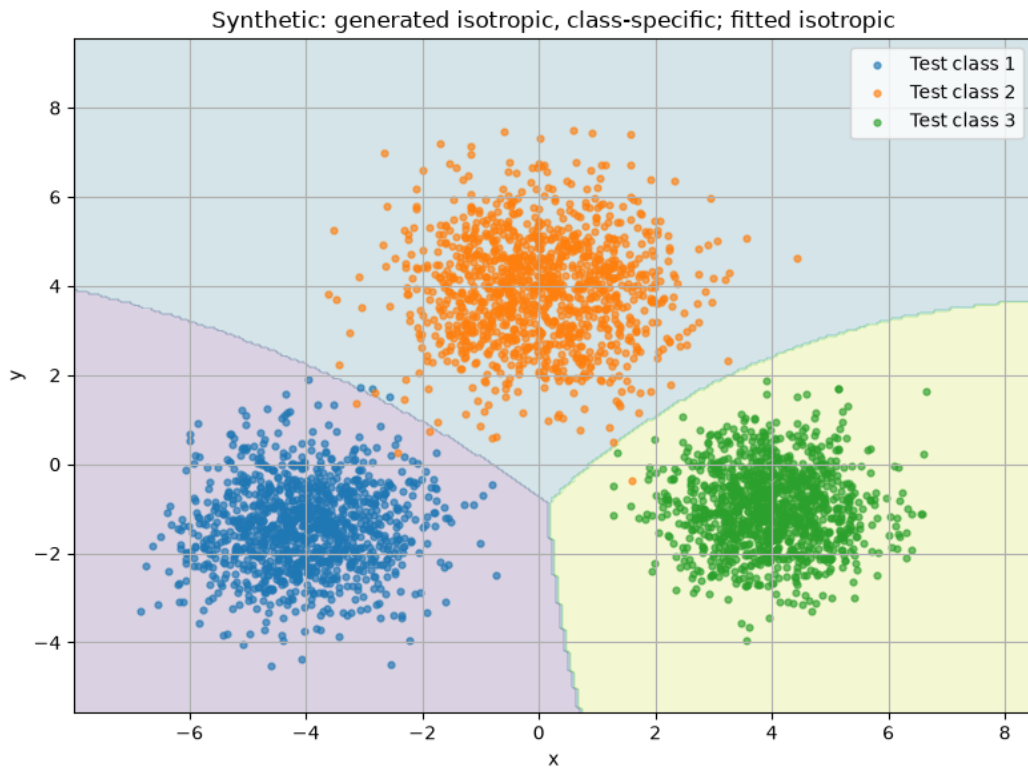


Figure 3: Q1 Figure 3: Confusion matrix plot



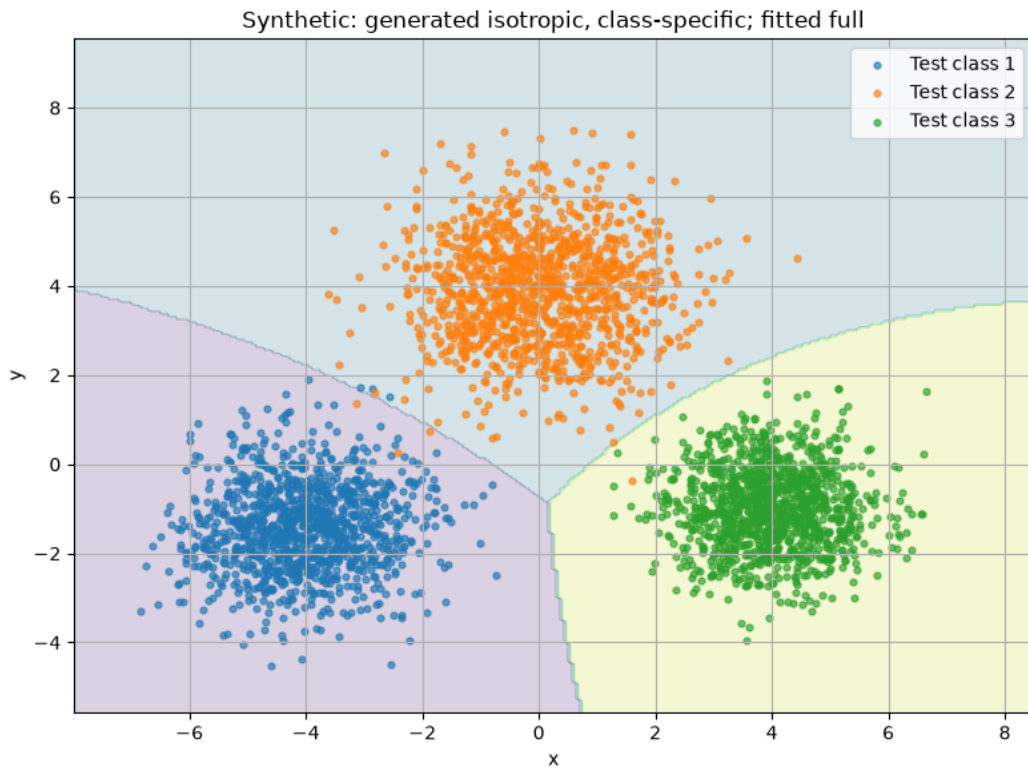


Figure 6: Q1 Figure 6: Confusion matrix plot

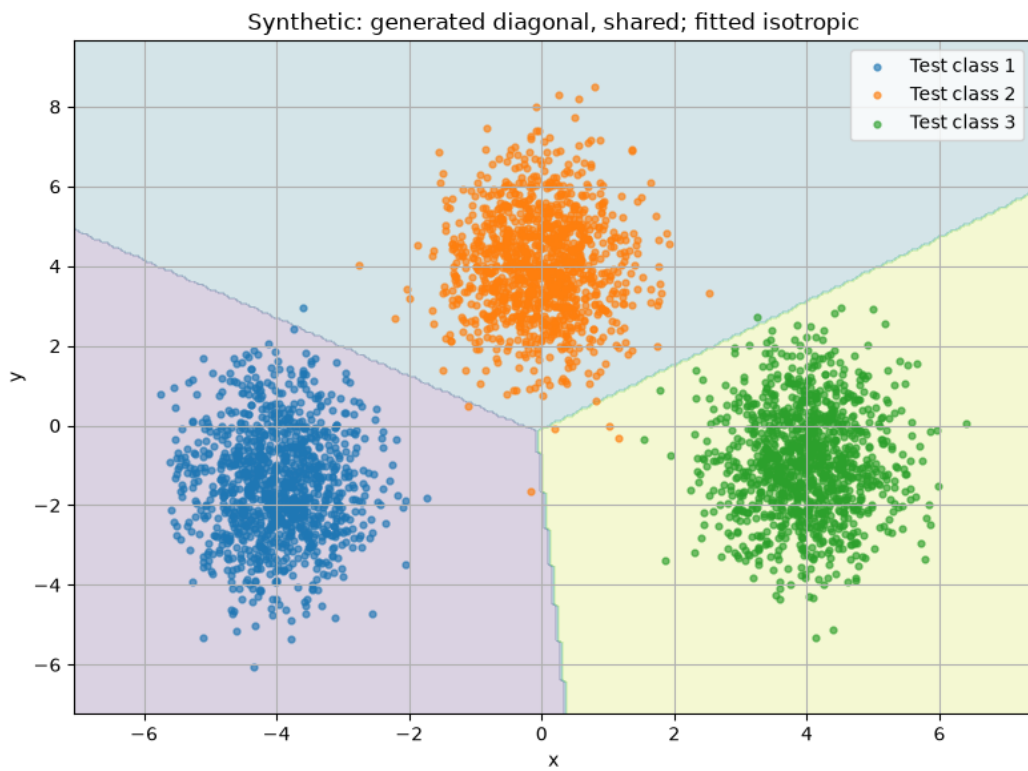


Figure 7: Q1 Figure 7: Confusion matrix plot

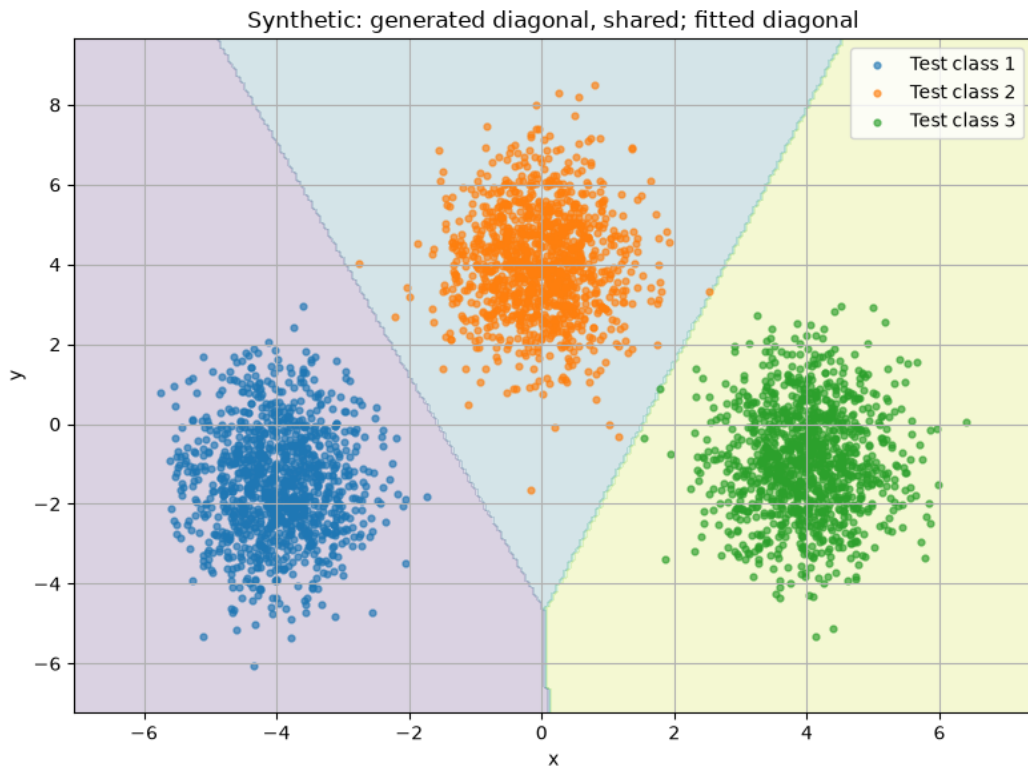


Figure 8: Q1 Figure 8: Confusion matrix plot

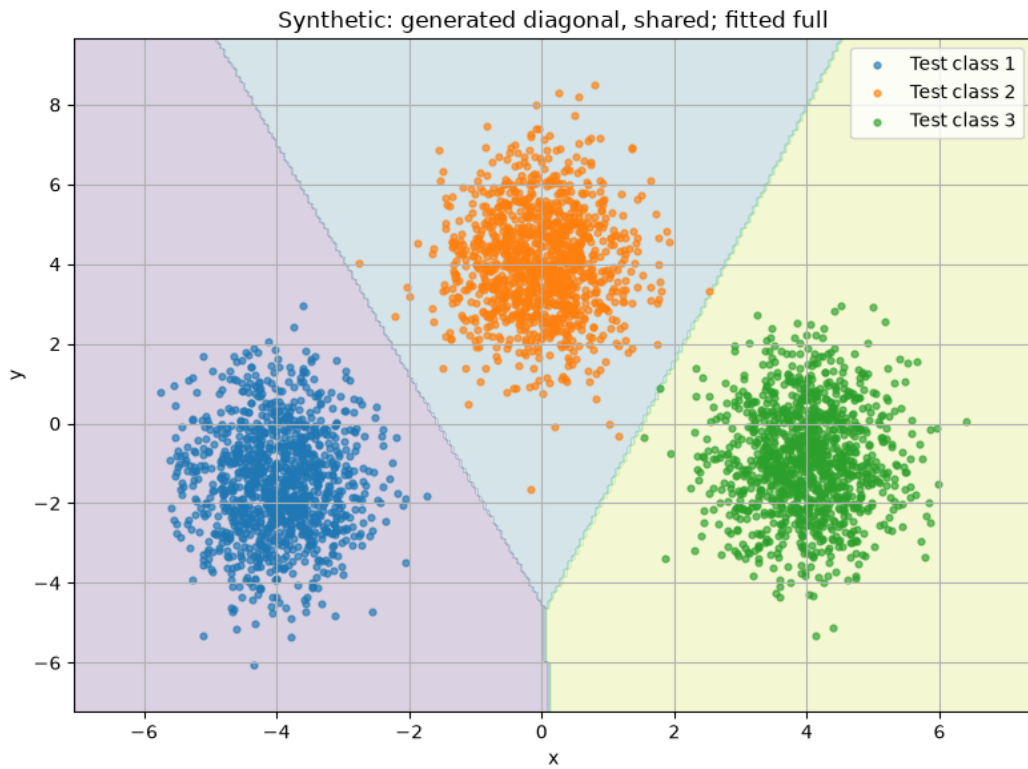


Figure 9: Q1 Figure 9: Confusion matrix plot

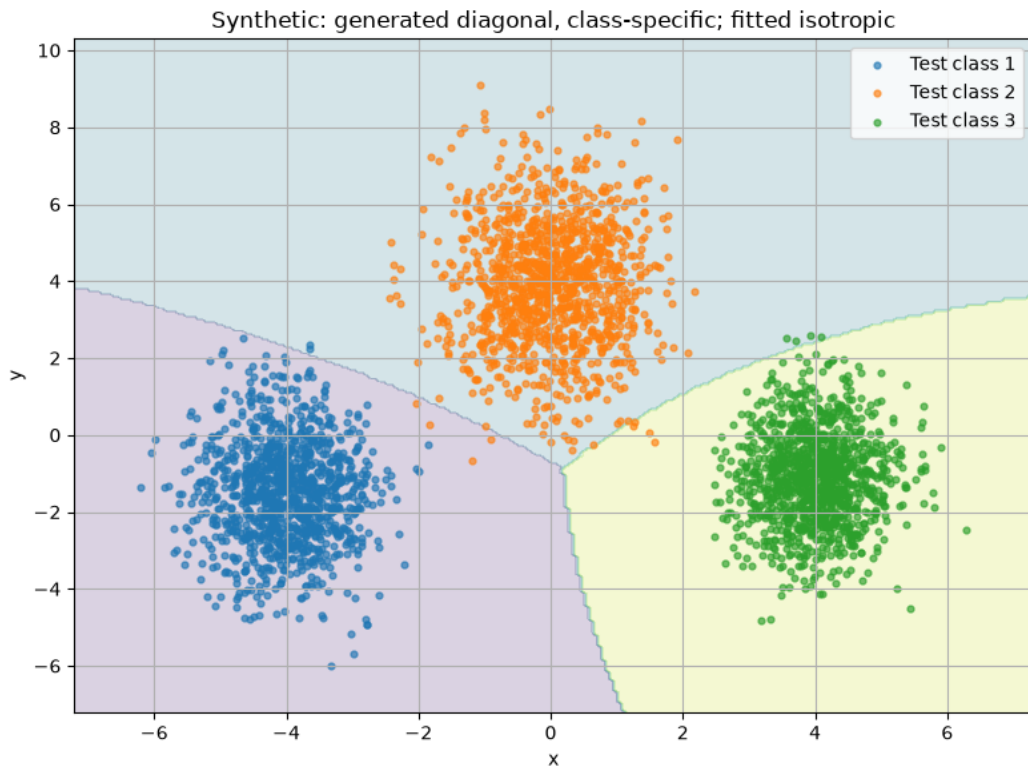


Figure 10: Q1 Figure 10: Confusion matrix plot

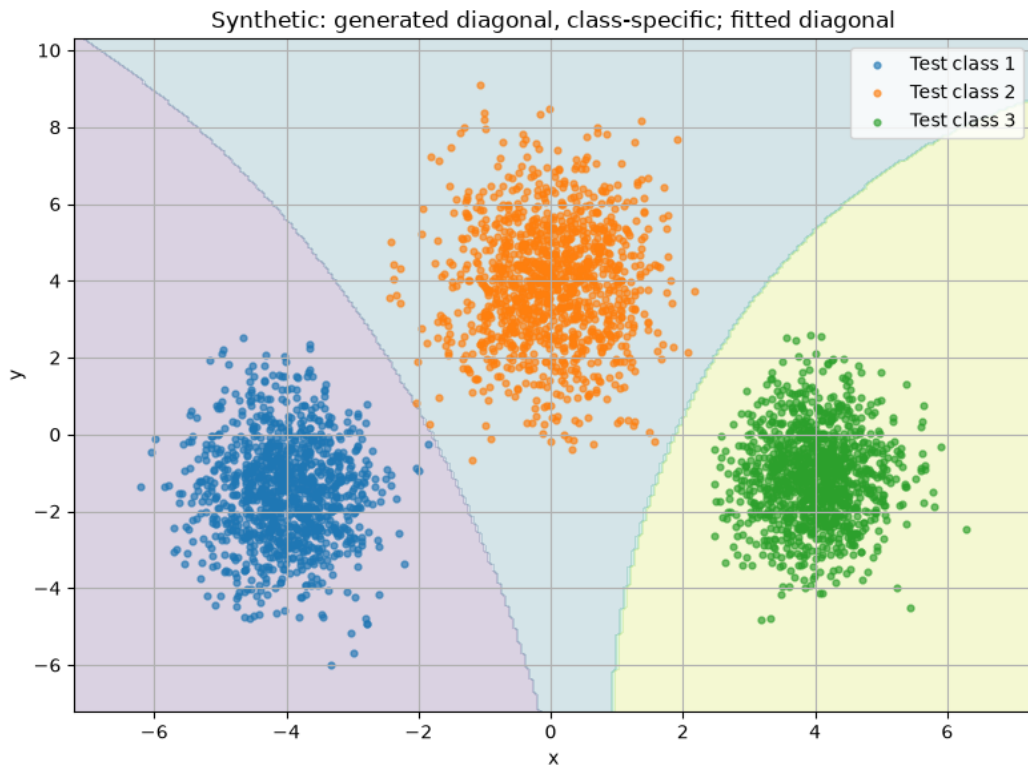


Figure 11: Q1 Figure 11: Confusion matrix plot

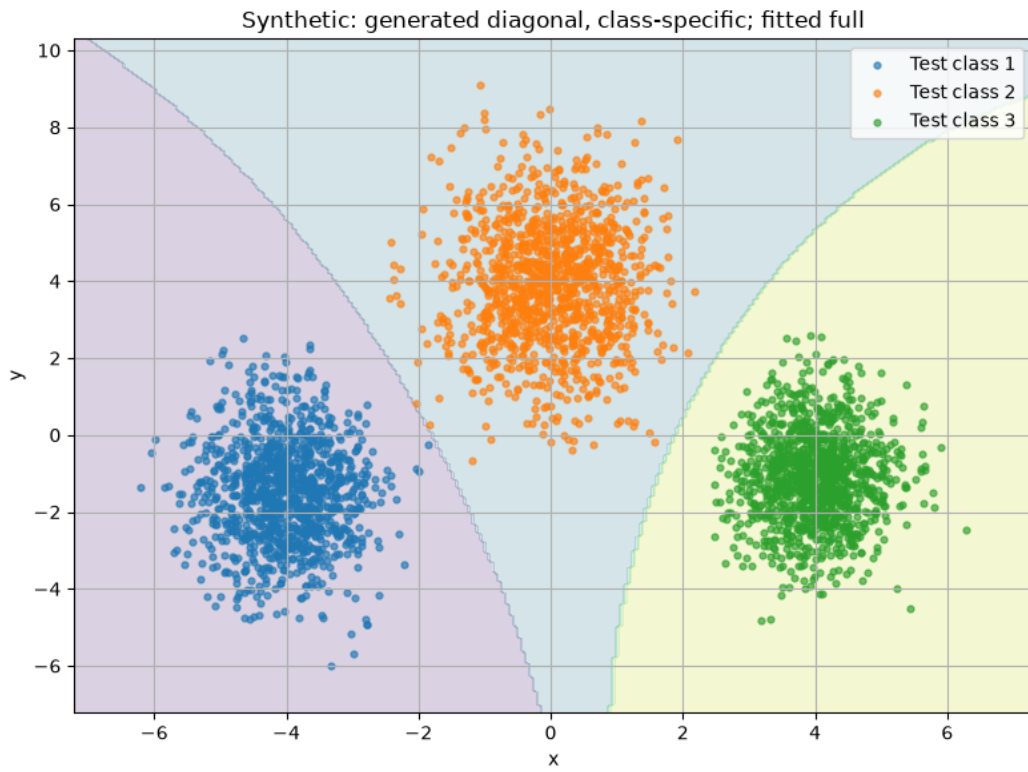


Figure 12: Q1 Figure 12: Confusion matrix plot

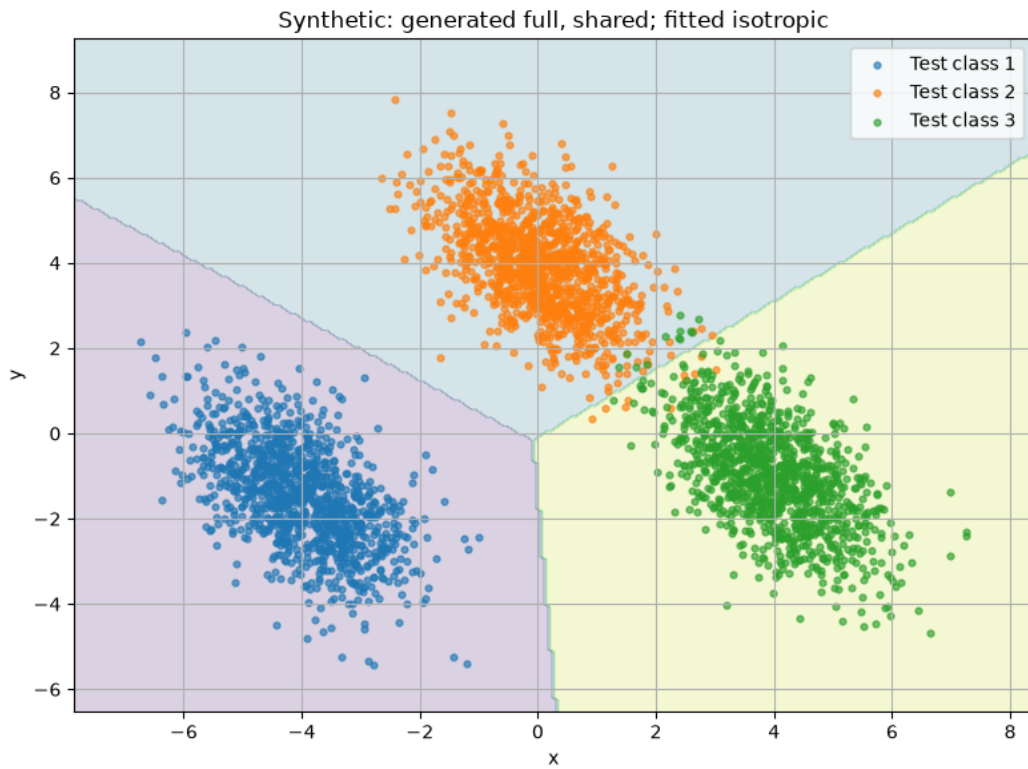


Figure 13: Q1 Figure 13: Confusion matrix plot

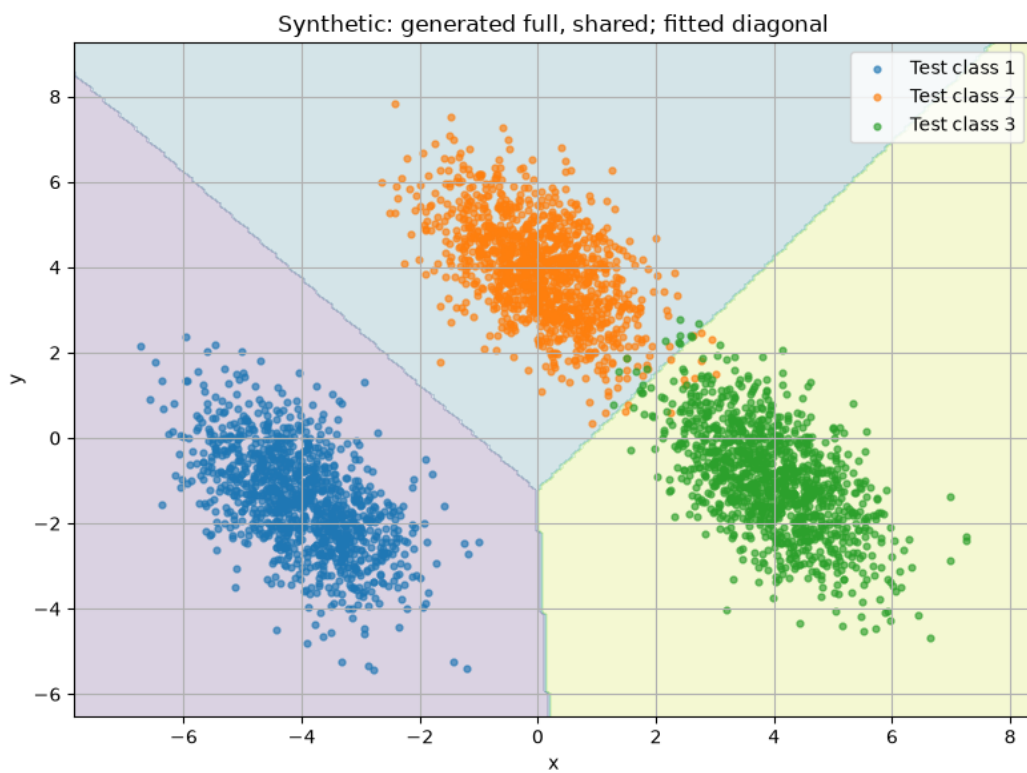


Figure 14: Q1 Figure 14: Confusion matrix plot

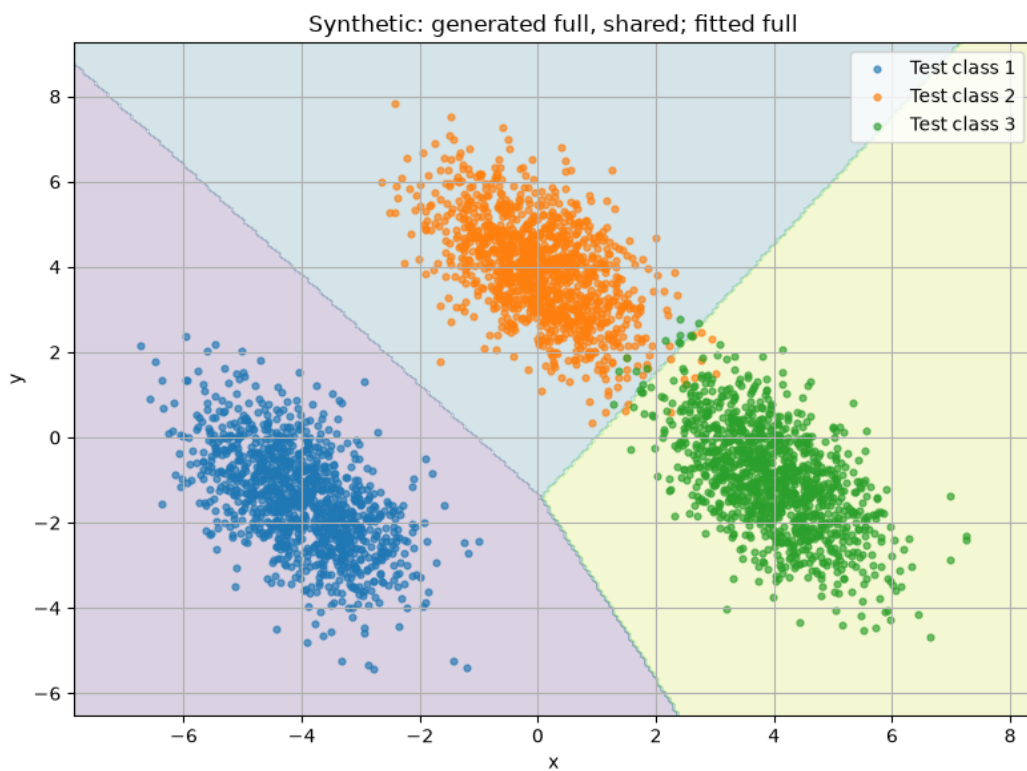


Figure 15: Q1 Figure 15: Confusion matrix plot

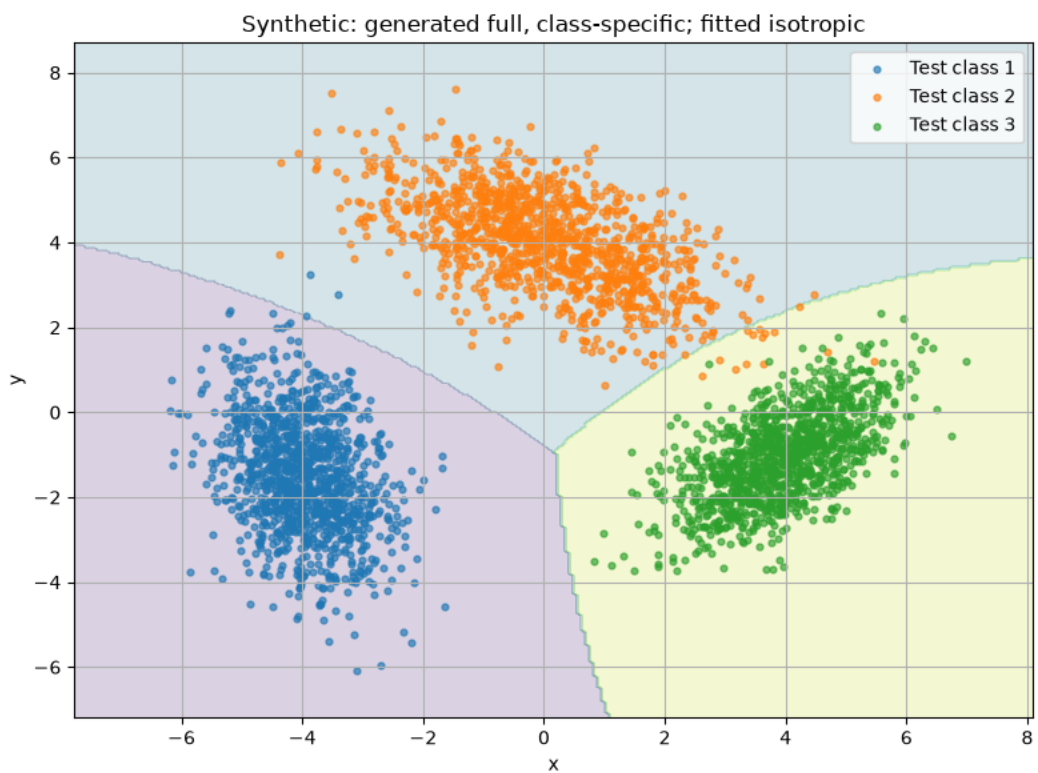


Figure 16: Q1 Figure 16: Confusion matrix plot

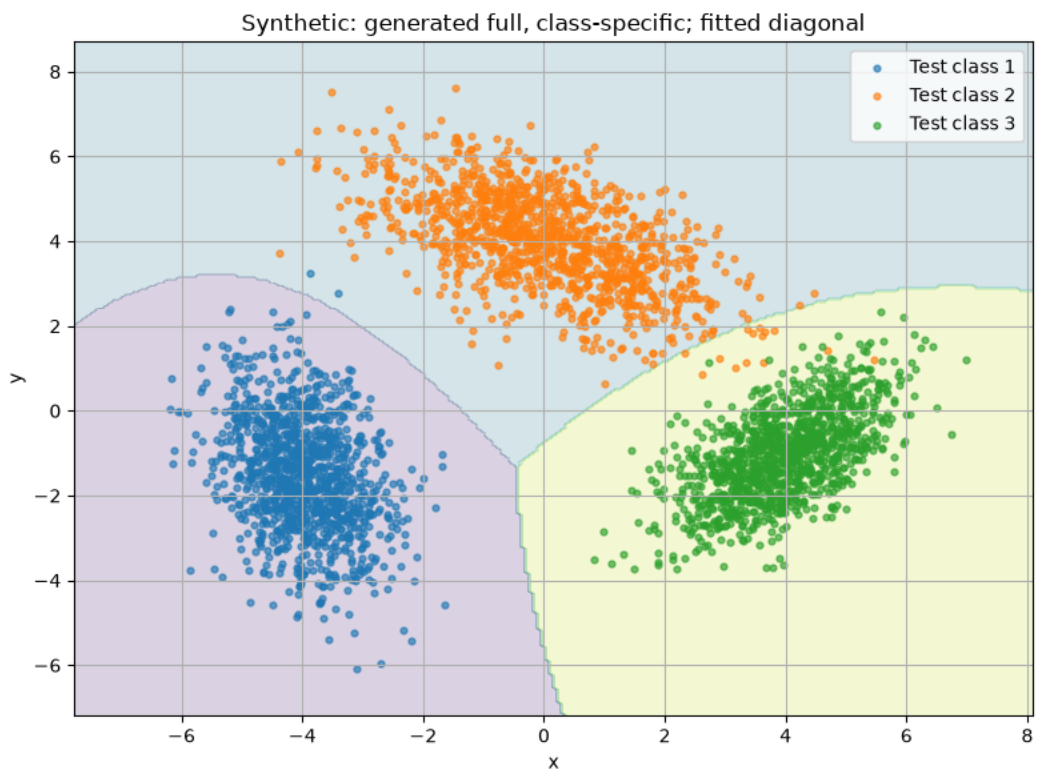


Figure 17: Q1 Figure 17: Confusion matrix plot

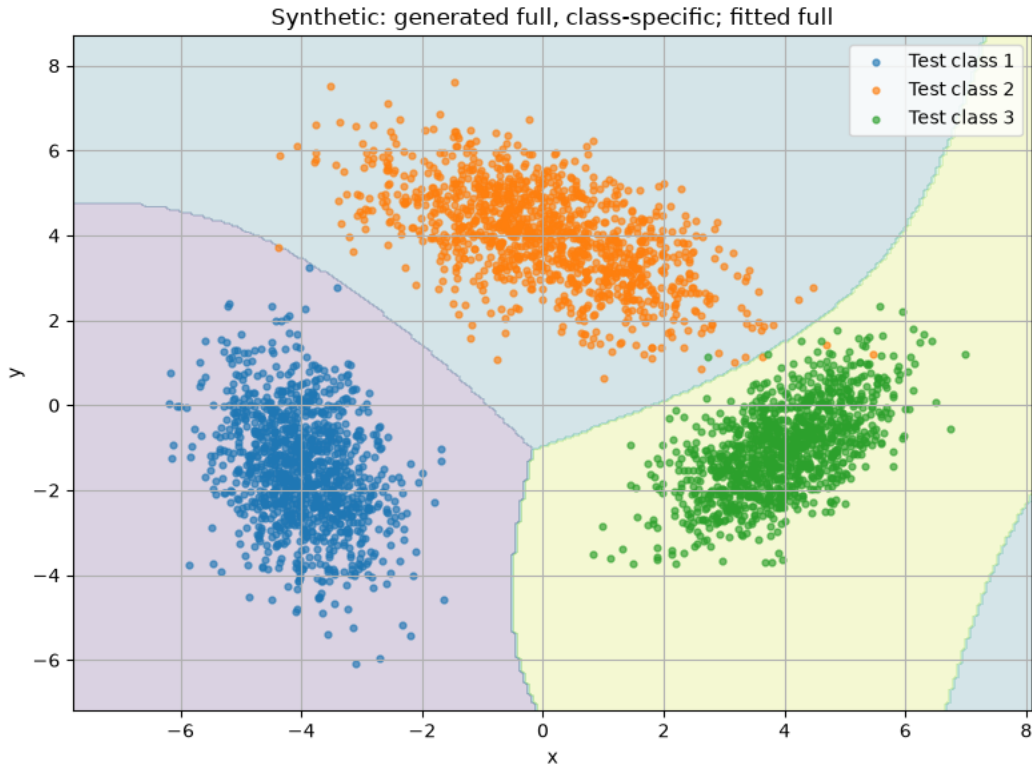


Figure 18: Q1 Figure 18: Confusion matrix plot

1.5 Observations and conclusion

Best-model accuracies exceed 99% because the synthetic classes are well separated. Covariance assumptions change the decision boundaries and class-wise errors. The confusion-matrix tables satisfy the assignment requirement.

2 Question 2: Real-world / Supplied 2-D Data

The uploaded training file has 1,050 rows (350 per class), and the held-out dev file has 300 rows (100 per class). The supplied split is 77.78%/22.22%.

2.1 Results

Model	Accuracy	Macro P	Macro R	Macro F1	Macro AUC
Isotropic	82.33%	0.833	0.823	0.824	0.9105
Diagonal	84.67%	0.852	0.847	0.848	0.9400
Full	84.33%	0.846	0.843	0.844	0.9407

2.2 Confusion matrices

Table 19: Real-world data; Isotropic covariance; accuracy 82.33%

Actual \ Predicted	1	2	3
Class 1	74	4	22
Class 2	2	87	11
Class 3	9	5	86

Table 20: Real-world data; Diagonal covariance; accuracy 84.67%

Actual \ Predicted	1	2	3
Class 1	78	2	20
Class 2	6	89	5
Class 3	12	1	87

Table 21: Real-world data; Full covariance; accuracy 84.33%

Actual \ Predicted	1	2	3
Class 1	78	2	20
Class 2	6	91	3
Class 3	13	3	84

2.3 Decision boundaries and confusion-matrix plots

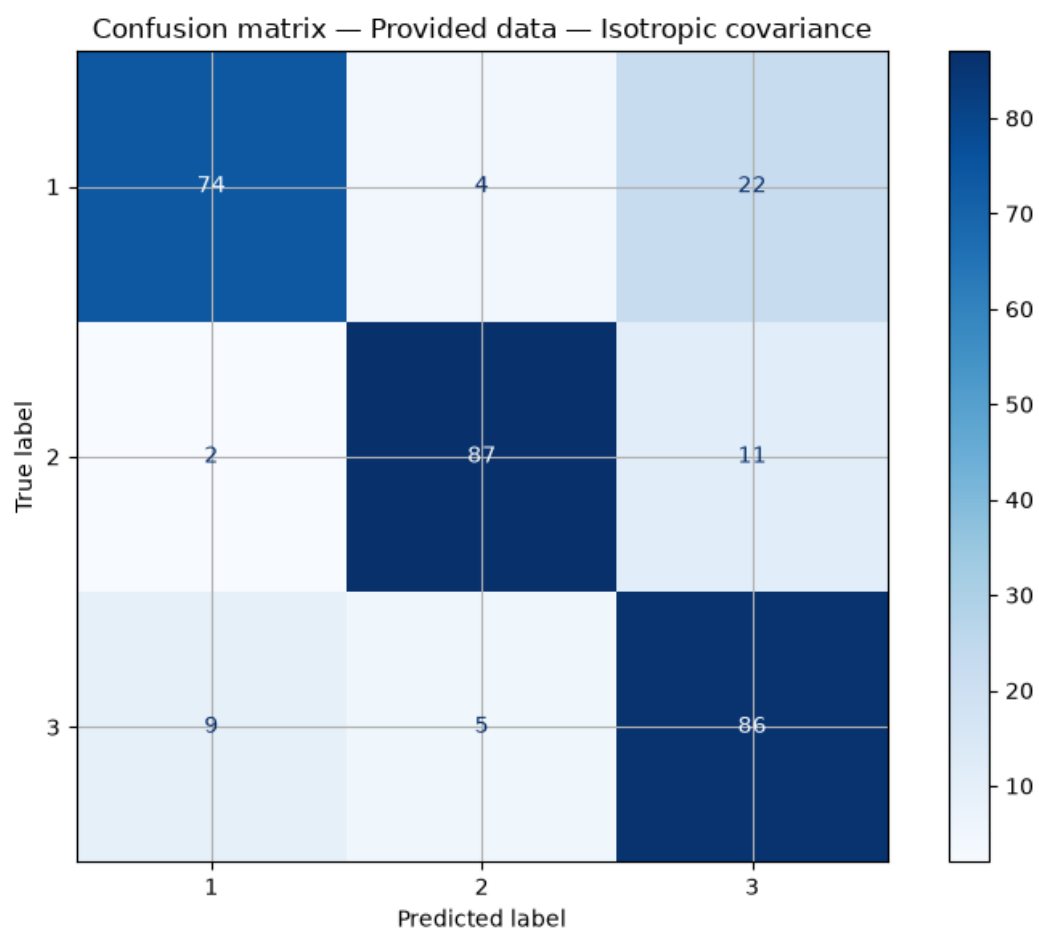


Figure 19: Q2 Figure 1: Confusion matrix plot

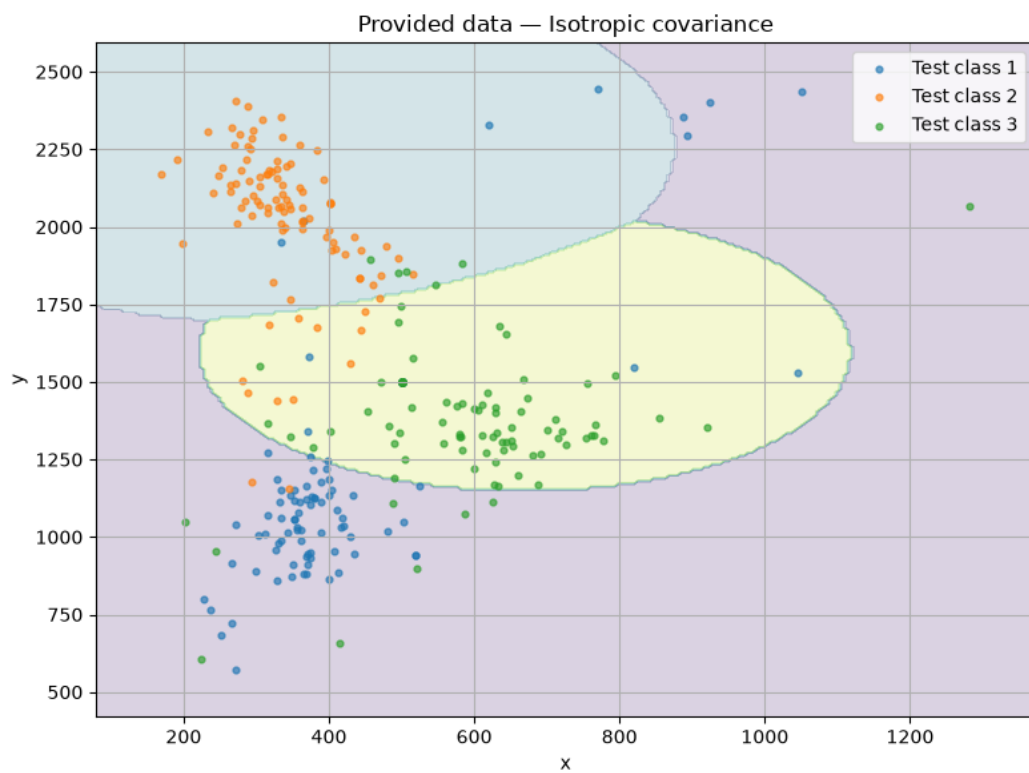


Figure 20: Q2 Figure 2: Confusion matrix plot

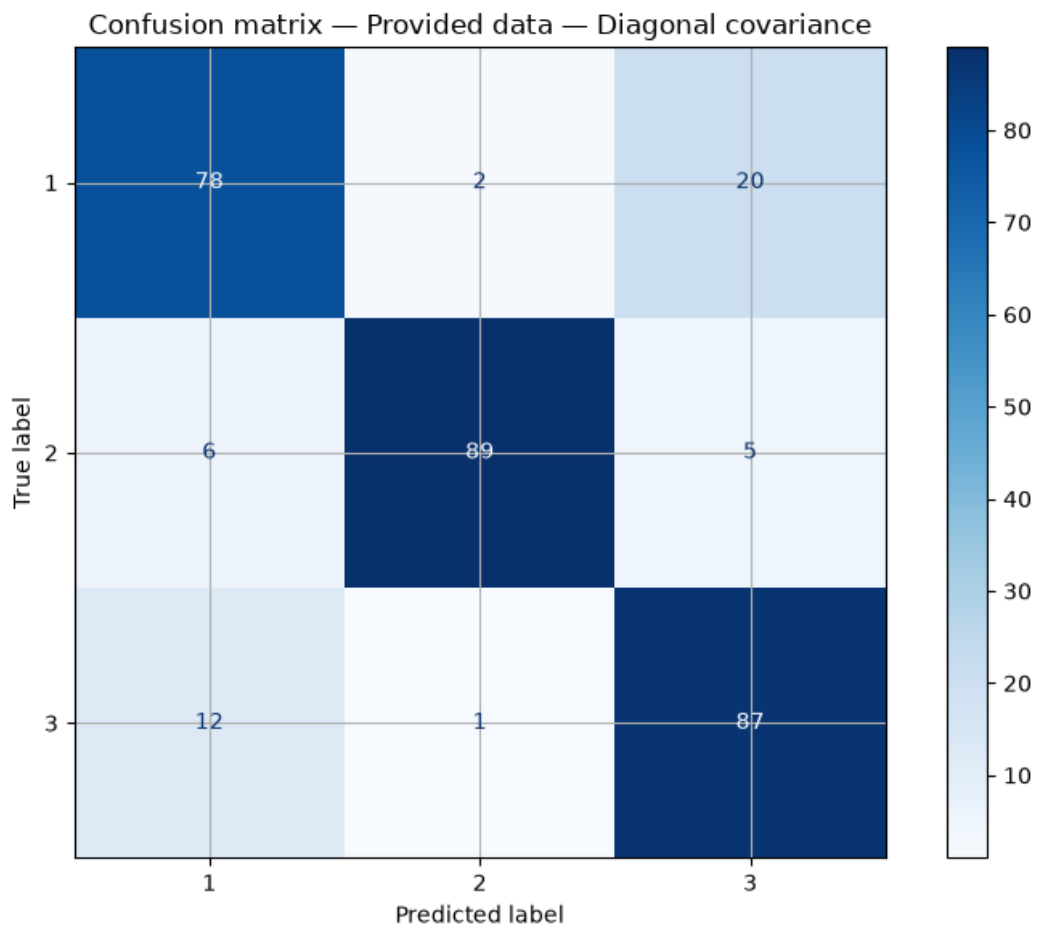


Figure 21: Q2 Figure 3: Confusion matrix plot

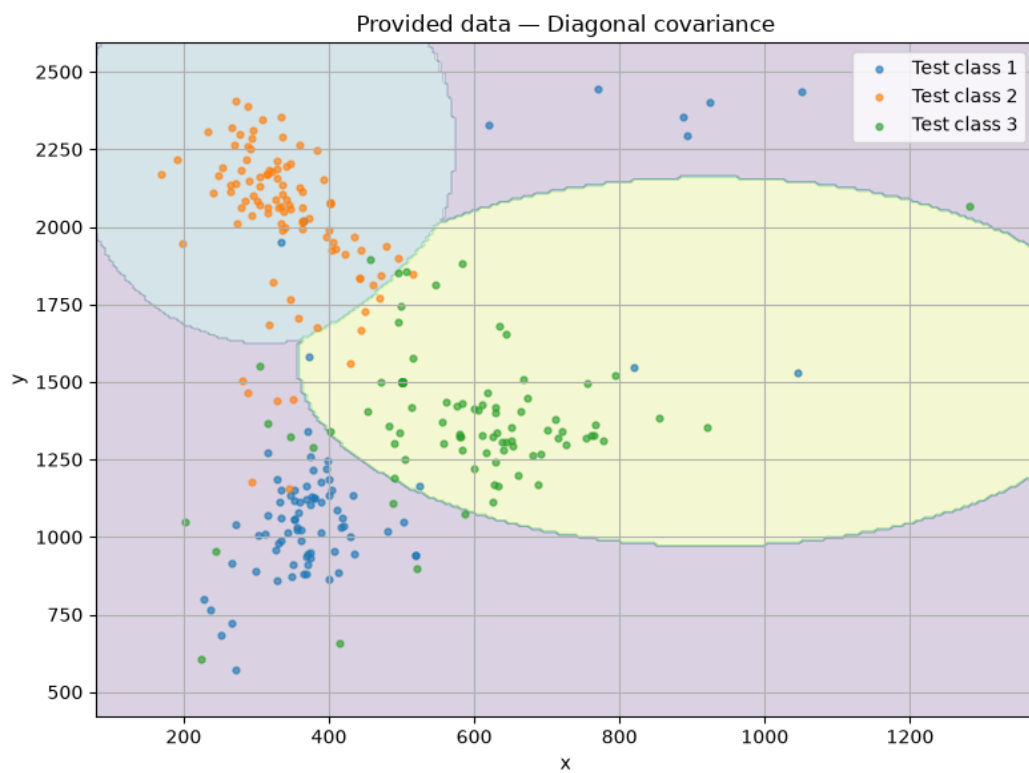


Figure 22: Q2 Figure 4: Confusion matrix plot

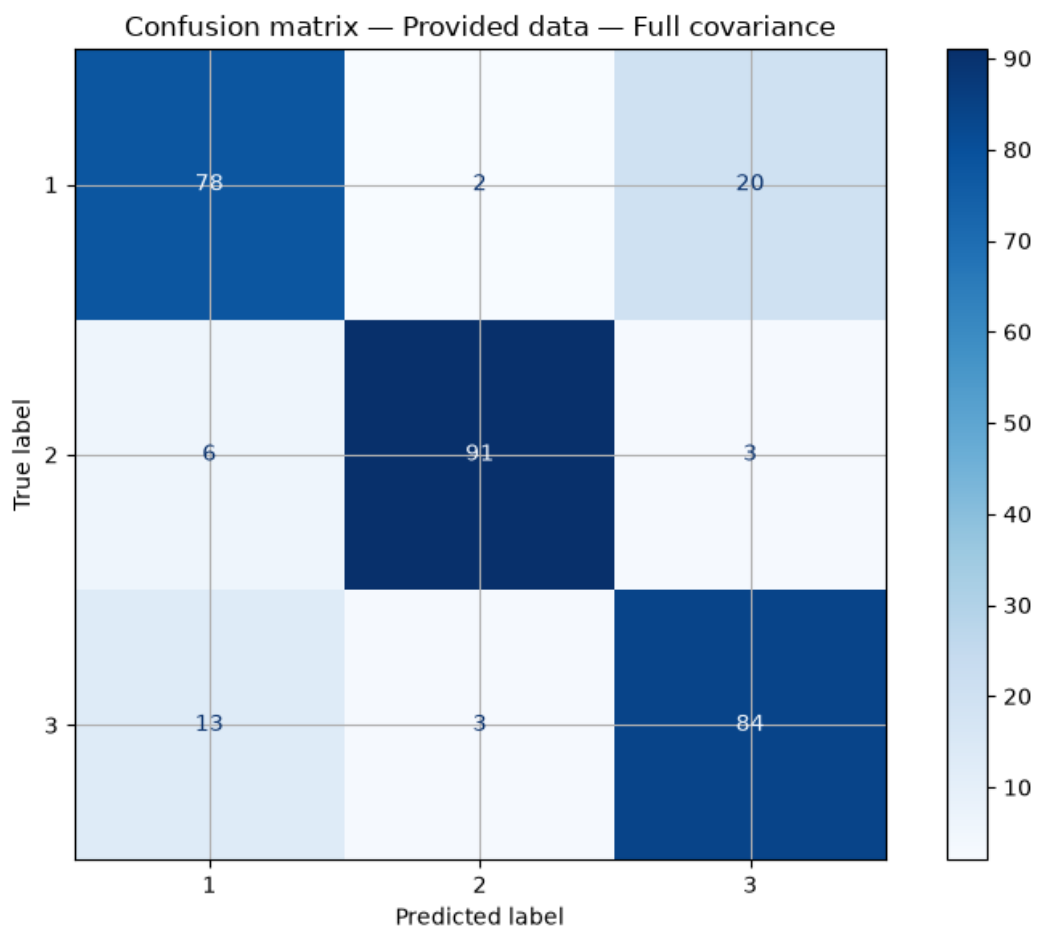


Figure 23: Q2 Figure 5: Confusion matrix plot

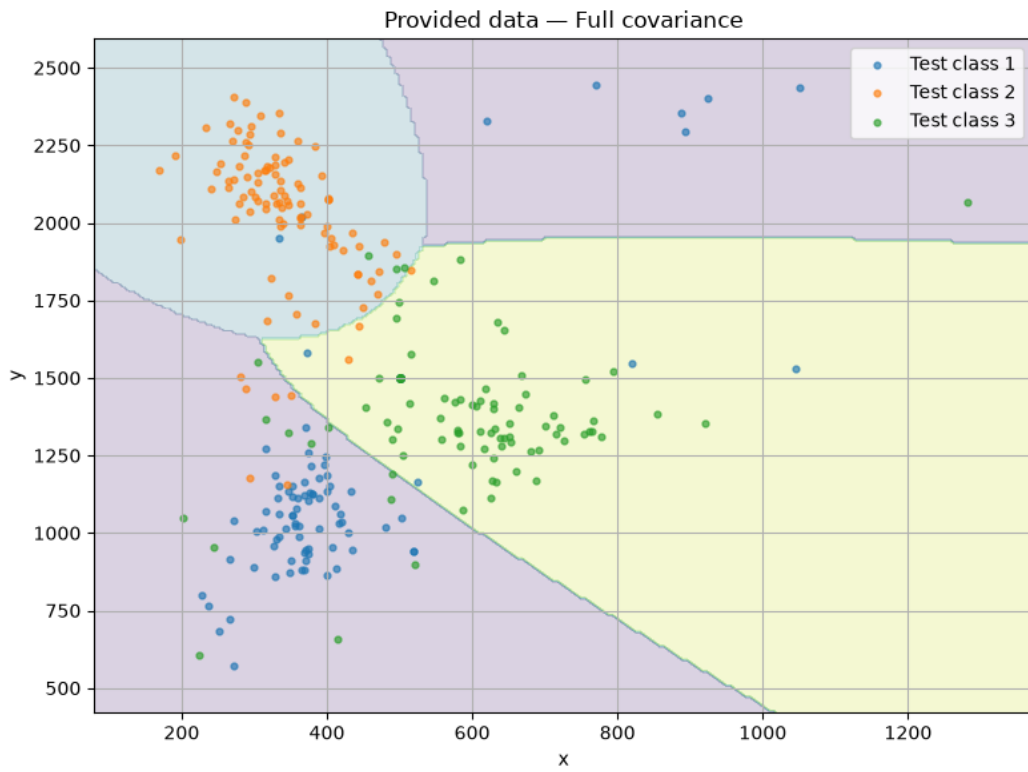


Figure 24: Q2 Figure 6: Confusion matrix plot

2.4 ROC curves

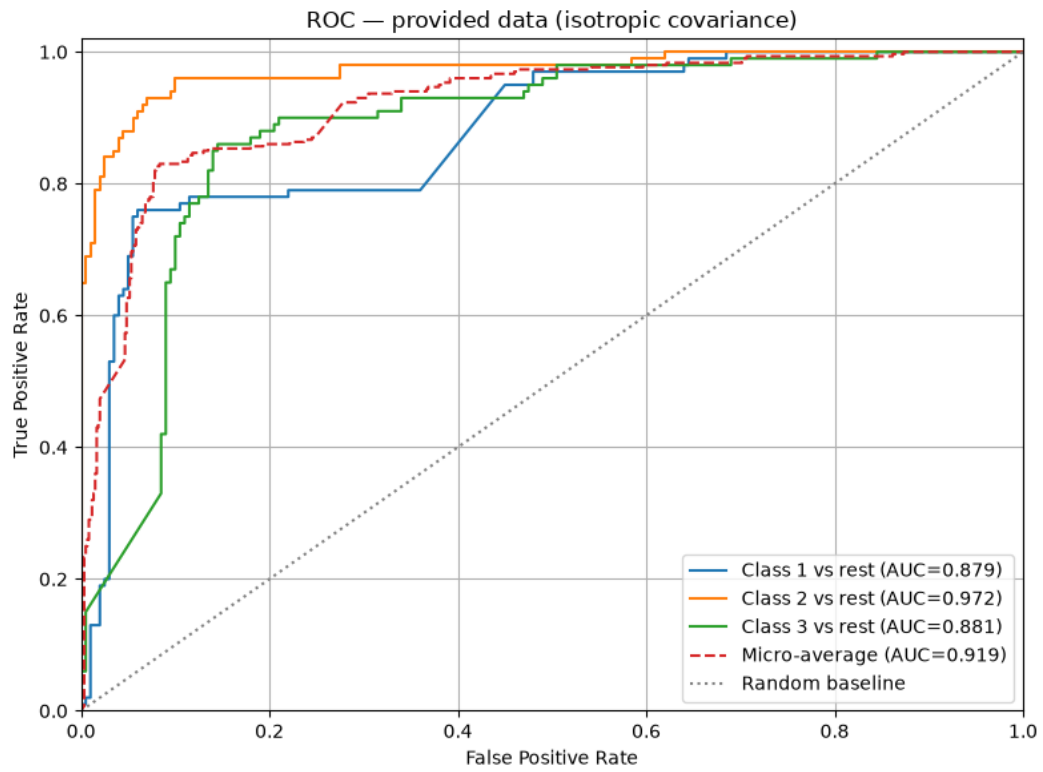


Figure 25: Q2 ROC Figure 1: ROC curve / AUC plot

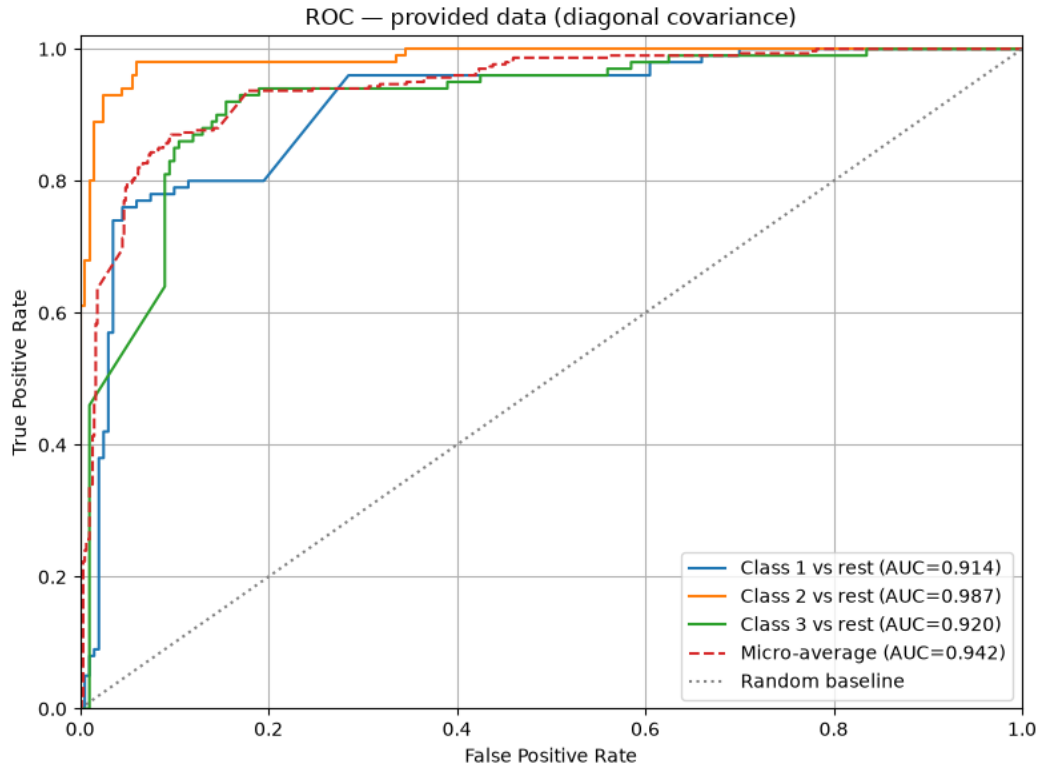


Figure 26: Q2 ROC Figure 2: ROC curve / AUC plot

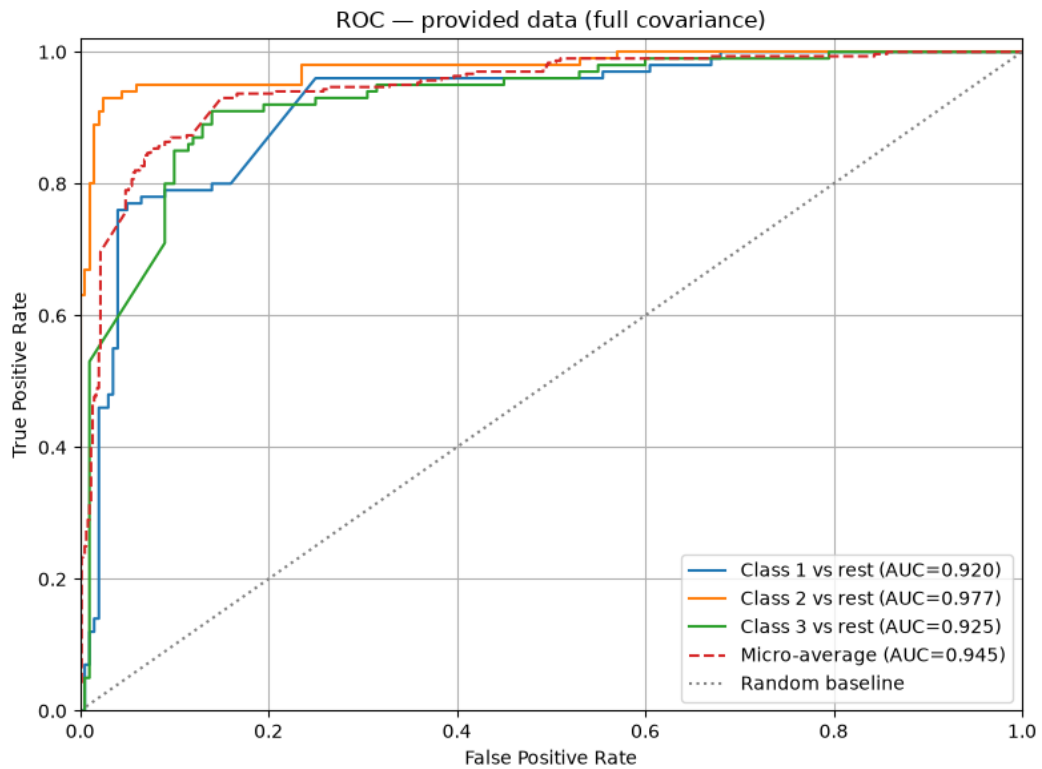


Figure 27: Q2 ROC Figure 3: ROC curve / AUC plot

2.5 Conclusion

For the supplied real-world dataset, diagonal covariance gives the highest held-out accuracy and macro-F1. Confusion matrices identify class-specific errors; decision boundaries and ROC curves

complement the metrics.

Overall Conclusion

Question 1 evaluates synthetic data and includes 18 confusion matrices. Question 2 evaluates supplied real-world data and includes its own confusion matrices, decision boundaries, and ROC curves.